

GTKWave: Look At Your Waveforms, Openly

ASHWIN GOPINATH

That opening line aside, this article is less about the pun and more about the information. If you are finding yourself on the search for a waveform viewer, then you are at the right page. In this article we cover a solution for the same.

GTKWave, developed by Tony Bybell, is a cross-platform waveform viewer that reads Verilog VHDL files and displays their forms. Though initially developed for Linux, it now operates on Microsoft Windows as well as Apple Mac OS X. As far as open source alternatives for waveform viewing go, it is the only software that runs on all three major operating systems.

GTKWave has been developed to handle debugging of large systems on chips (SoCs), as it is run on a workstation with sufficient amount of physical memory to handle the task. Commonly-used file formats like LXT2 and VZT have been developed to handle real-world designs, whereas formats like AET2 and FST have been designed to handle extremely-large designs.

Components of GTKWave

GTKWave tool has two separate but interlocking tools, that is, gtkwave viewer and rtlbrowse. There are a bunch of applications under the hood that assist in the working while working themselves under independent situations.

gtkwave is the name of the primary tool under use, which helps with all major visualisations. If a full simulation dump is input, then it can help in data visualisation of analogue and digital data, while

TABLE I VARIOUS FORMATS FOR GTKWAVE		
Format	Abbreviation for	What it is
VCD	Value change dump	Industry standard format generated by most simulators; slow but ubiquitous
LXT	interLacedXtensible Trace	Optimised for use; processes faster but used only by GTKWave and a handful others
LXT2	interLacedXtensible Trace v2	Block based variant; allows for greater compression; supported additionally by Icarus
VZT	Verilog Zipped Trace	Cousin of LXT but file sizes much smaller; slowest write performance but fast on microprocessors
GHW	GHDH Wave file	GHDH wave file; nine-state file format

allowing for search operations, temporal manipulations and save partial results. For hard copies, you can export the results in PostScript and FrameMaker output.

rtlbrowse is used to view and navigate through RTL source code that has been passed and processed into a stems file by vermin—another helper application. It allows for viewing of RTL at file and module levels. To annotate the code at this level, invoke it by gtkwave. Functions like file conversion, RTL parsing and other data-manipulation operations are performed by the aforementioned helper applications.

Multiple signal processing

GTKWave is designed to process a host of signals at once. As such, it is home to three different signal-searching modes, namely, signal search, hierarchy and tree.

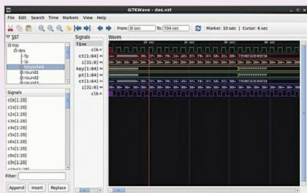


Fig. 1: GTKWave running under Linux

Signal search. In this mode, the search requester accepts a search string as a regular expression. If a signal is found matching the string, it is listed in Matches section. These can be controlled and manipulated by either dragging one or more into gtkwave viewer. After selecting, there are options to append, replace or insert, as required.

Hierarchy search. Hierarchy search, as the name suggests, provides a view of the hierarchy in a format similar to the current working directory. In essence, it is the same as the file explorer for any operating system. It contains the